

Chapter 6 – Network Configuration & Remote Administration

Objective

Configure Ubuntu Server networking for the Wazuh Home Lab by:

- Configuring a static IP address.
 - Using two network adapters.
 - Managing the server remotely using SSH.
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1. Why Does Wazuh Need a Static IP?

A **Static IP Address** is a permanent IP address that does not change after reboot.

Example:

192.168.56.10

Why is it important?

- Wazuh Agents communicate with the Wazuh Manager using its IP address.
 - If the Manager's IP changes, agents lose communication.
 - Servers should always use static IP addresses.
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2. DHCP vs Static IP

DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP addresses.
- IP addresses may change after reboot.

Example:

Today

192.168.56.102

Tomorrow

192.168.56.105

Advantages

- Automatic configuration
- Easy to manage

Disadvantages

- IP address is not permanent.
-

Static IP

- Configured manually.
- Remains the same until changed.

Example:

192.168.56.10

Advantages

- Stable
 - Predictable
 - Recommended for servers
-

3. Why Two Network Adapters?

Our Ubuntu Server uses two interfaces.

Adapter 1 – NAT

Purpose:

- Internet Access
- Ubuntu Updates

- Download Wazuh Packages

Interface

enp0s3

Example IP

10.0.2.15

Adapter 2 – Host-Only

Purpose:

- Private communication between VMs
- Wazuh Agent communication
- Kali attack simulation

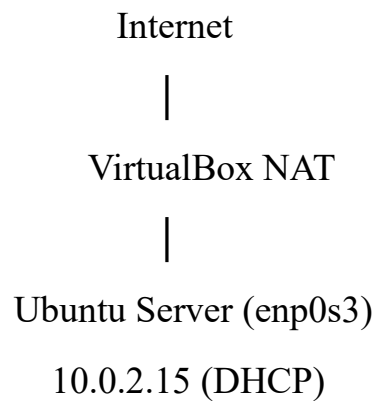
Interface

enp0s8

Static IP

192.168.56.10

4. Lab Network Design



Host-Only Network

Ubuntu Server

192.168.56.10



Windows



Kali Linux

5. Netplan

Netplan is Ubuntu's network configuration utility.

Used to configure:

- Static IP
- DHCP
- Gateway
- DNS
- Network Interfaces

Configuration files are stored in:

/etc/netplan/

Example:

/etc/netplan/00-installer-config.yaml

6. YAML

YAML stands for:

YAML Ain't Markup Language

It is a human-readable configuration format used by tools such as:

- Netplan
- Docker Compose
- Kubernetes
- Ansible

Important Rule

YAML depends on **spaces**, not tabs.

Incorrect indentation causes configuration errors.

7. Backup the Configuration

Always create a backup before editing system configuration files.

Command

```
sudo cp /etc/netplan/00-installer-config.yaml \  
/etc/netplan/00-installer-config.yaml.backup
```

Purpose:

Restore the original configuration if needed.

8. Edit Netplan

Command

```
sudo nano /etc/netplan/00-installer-config.yaml
```

Purpose:

Edit the network configuration.

9. Static IP Configuration

Example:

network:

version: 2

ethernets:

enp0s3:

dhcp4: true

dhcp6: true

enp0s8:

dhcp4: false

addresses:

- 192.168.56.10/24

Explanation

Setting	Purpose
network:	Starts the network configuration
version: 2	Netplan configuration version

ethernets:	Defines Ethernet interfaces
enp0s3	NAT interface
dhcp4: true	Automatically obtain IPv4 address
enp0s8	Host-Only interface
dhcp4: false	Disable DHCP
addresses:	Assign static IP
/24	Subnet mask (255.255.255.0)

10. Validate Configuration

Command

```
sudo netplan generate
```

Purpose:

Checks the YAML syntax before applying changes.

If there is **no output**, the configuration is valid.

11. Test Configuration

Command

```
sudo netplan try
```

Purpose:

- Temporarily applies the configuration.
 - Automatically rolls back if there is a problem.
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12. Apply Configuration

Command

```
sudo netplan apply
```

Purpose:

Applies the network configuration permanently.

Difference Between netplan try and netplan apply

netplan try	netplan apply
Tests configuration	Applies configuration permanently
Automatically rolls back on failure	No automatic rollback
Recommended before applying	Used after successful testing

13. Verify Network

Display IP Addresses

```
ip addr
```

Display Routing Table

```
ip route
```

Check Internet

```
ping google.com
```

Display Hostname

```
hostnamectl
```

14. Remote Administration Using SSH

What is SSH?

SSH (Secure Shell) is a secure protocol used to remotely manage Linux servers over an encrypted connection.

Instead of working directly in the VirtualBox console, administrators connect to the server from another computer.

Why Use SSH?

- Secure encrypted communication
 - Remote administration
 - Easy copy and paste
 - Faster than using the VM console
 - Standard method used by Linux administrators and SOC analysts
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Verify SSH Service

```
sudo systemctl status ssh
```

If SSH is not running:

```
sudo systemctl enable ssh
```

```
sudo systemctl start ssh
```

Connect from Windows

Open:

- Windows PowerShell
- Windows Terminal

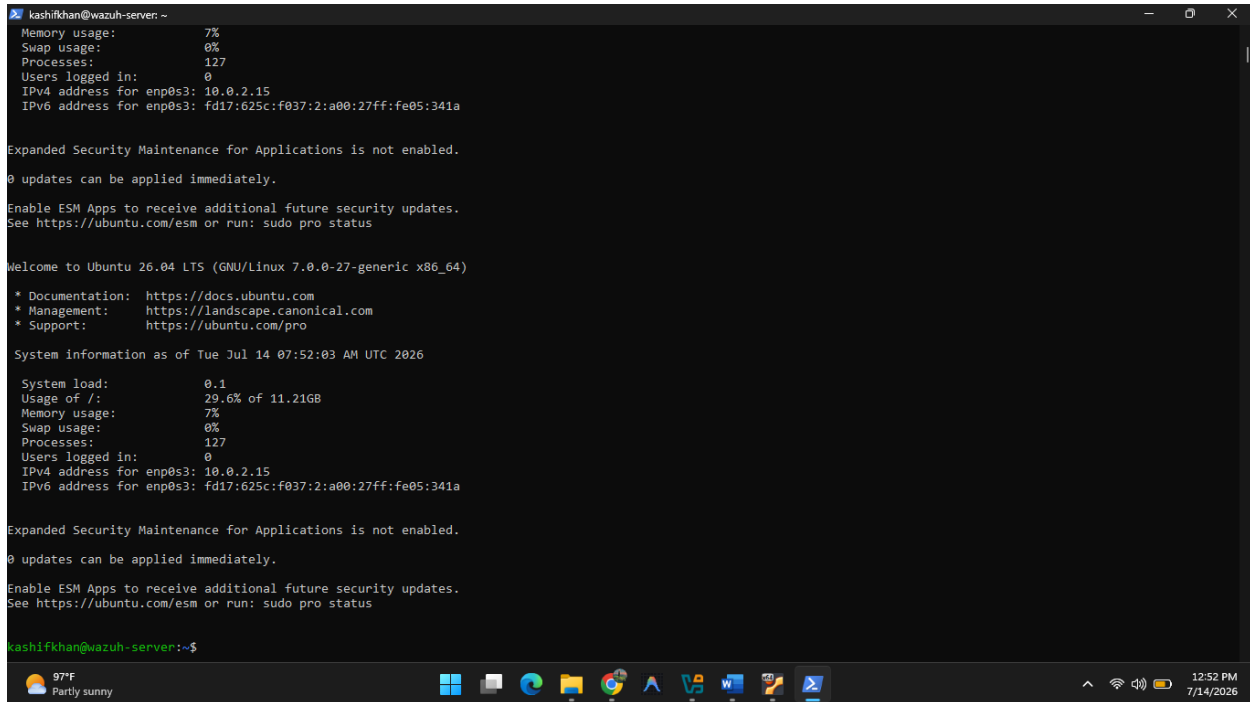
Command:

```
ssh username@ip-address
```

Example:

ssh kashifkhan@192.168.56.10

After connecting, enter your Ubuntu password.



```
kashifkhan@wazuh-server:~  
Memory usage:      7%  
Swap usage:        0%  
Processes:         127  
Users logged in:   0  
IPv4 address for enp0s3: 10.0.2.15  
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe05:341a  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-27-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Tue Jul 14 07:52:03 AM UTC 2026  
  
System load:        0.1  
Usage of /:         29.6% of 11.21GB  
Memory usage:       7%  
Swap usage:         0%  
Processes:         127  
Users logged in:    0  
IPv4 address for enp0s3: 10.0.2.15  
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe05:341a  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
kashifkhan@wazuh-server:~$
```

Common Windows SSH Issue

Problem

ssh : The term 'ssh' is not recognized...

Cause

The OpenSSH Client is installed but the OpenSSH directory is not available in the Windows PATH.

Verify

where.exe ssh

Temporary Fix

\$env:Path += ";C:\Windows\System32\OpenSSH"

Permanent Fix

Add the following directory to the Windows **Path** environment variable:

C:\Windows\System32\OpenSSH

15. Final Network

Interface	Purpose	IP Address
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enp0s3	NAT	10.0.2.15 (DHCP)
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enp0s8	Host-Only	192.168.56.10 (Static)
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Chapter 6 Summary

- Wazuh Manager requires a **Static IP Address**.
 - Use **NAT** for Internet access.
 - Use **Host-Only** for communication with lab machines.
 - Ubuntu uses **Netplan** to manage networking.
 - Validate configuration using `netplan generate`.
 - Test using `netplan try`.
 - Apply using `netplan apply`.
 - Verify using `ip addr`, `ip route`, and `ping`.
 - Manage Ubuntu remotely using **SSH** from Windows.
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Commands Cheat Sheet

Command	Purpose
ip addr	Display IP addresses

<code>ip route</code>	Display routing table
<code>hostnamectl</code>	Display hostname
<code>sudo nano /etc/netplan/00-installer-config.yaml</code>	Edit Netplan
<code>sudo cpbackup</code>	Backup Netplan file
<code>sudo netplan generate</code>	Validate YAML syntax
<code>sudo netplan try</code>	Test configuration
<code>sudo netplan apply</code>	Apply configuration
<code>ping google.com</code>	Verify Internet connectivity
<code>sudo systemctl status ssh</code>	Check SSH service
<code>ssh username@ip-address</code>	Connect remotely using SSH
